(2) Spray a small stream of water (Spray Bottle) at the suspected leak area.
(3) If engine RPM'S change, the area of the suspected leak has been found.
(4) Repair as required.

REMOVAL

(1) Disconnect negative cable from battery.
(2) Disconnect connector from inlet air temperature sensor.
(3) Disconnect air intake tube at throttle body and remove upper air cleaner housing.
(4) Disconnect connector from throttle position sensor (TPS).
(5) Disconnect connector from idle air control (IAC) motor.
(6) Disconnect connector from MAP sensor.
(7) Remove vacuum lines for purge solenoid and PCV valve at intake manifold.
(8) Remove vacuum lines for power brake booster, LDP, EGR transducer, and speed control vacuum reservoir (if equipped) at intake manifold fittings.
(9) Disconnect throttle, speed control (if equipped), and transaxle control (if equipped) and cables from throttle lever and bracket. (Refer to 14 - FUEL SYSTEM/FUEL INJECTION/THROTTLE CONTROL CABLE - REMOVAL)
(10) Perform fuel system pressure release procedure **before attempting any repairs.** (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY - STANDARD PROCEDURE)
(11) Disconnect fuel line. (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY/QUICK CONNECT FITTING - STANDARD PROCEDURE)
(12) Disconnect coolant temperature sensor/fuel injector wire harness connector.
(13) Disconnect fuel injector harness.
(14) Remove intake manifold to cylinder head fasteners.
(15) Remove the manifold from engine.

CAUTION: Cover intake manifold openings to prevent foreign material from entering engine.

(16) Inspect the manifold. (Refer to 9 - ENGINE/MANIFOLDS/INTAKE MANIFOLD - INSPECTION)

INSPECTION

(1) Check manifold surfaces for flatness with straight edge. Surface must be flat within 0.15 mm per 300 mm (0.006 in. per foot) of manifold length.
(2) Inspect manifold for cracks or distortion. Replace manifold if necessary.

INSTALLATION

(1) Clean manifold sealing surfaces.
(2) Install new manifold to cylinder head seals.
(3) Install manifold to head.
(4) Install and tighten intake manifold fasteners to 28 N·m (250 in. lbs.) in the sequence shown (Fig. 92). Repeat procedure until all bolts are at specified torque.

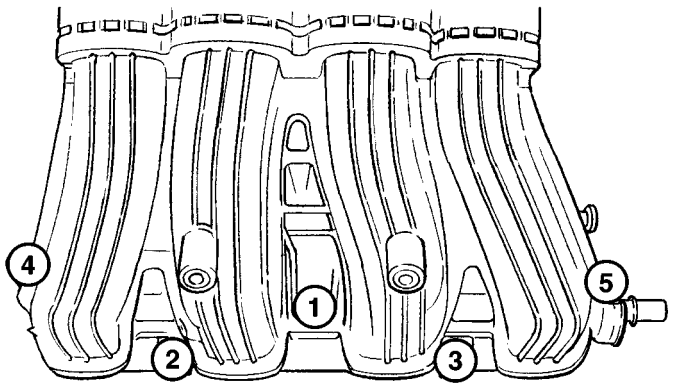

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Fig. 92 LOWER INTAKE MANIFOLD TIGHTENING SEQUENCE

(5) Install throttle cables in bracket.
(6) Connect throttle, speed control, (if equipped), cables to throttle lever.
(7) Connect vacuum lines for power brake booster, LDP, EGR transducer, and speed control vacuum reservoir (if equipped) at upper intake manifold fittings.
(8) Connect vacuum lines for purge solenoid and PCV valve.
(9) Connect electrical connectors for MAP sensor, throttle position sensor (TPS), and idle air control (IAC) motor.
(10) Connect the fuel line. (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY/QUICK CONNECT FITTING - STANDARD PROCEDURE)
(11) Connect coolant temperature sensor/fuel injector wiring harness electrical connector.
(12) Install the air cleaner housing and air intake tube to throttle body.
(13) Connect inlet air temperature sensor connector.
(14) Connect negative cable to battery.

EXHAUST MANIFOLD

DESCRIPTION

The exhaust manifold is made of Hi-Silicone Moly nodular cast iron for strength and high temperatures. The manifold attaches to the cylinder head.